SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Peer Review

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Peer Review
Weightage:	30% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	A.Likithasri	**Reg. No.:**	192511003
Section / Batch:	1	**Date:**	22/07/2026

Code of Conduct

I A.Likithasri(192511003)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Likithasri.A

Instructions

- This assessment contains peer review with individual presentation. Total: 100 Marks.
- Students should answer technical questions from peers and faculty.
- No negative marking. Unanswered questions carry zero marks.
- Students can use PowerPoint/Whiteboard/Live Coding/Demonstration for presentation.

Section / Topic	Focus	Q Nos	Marks
Linear Data Structure	Linked List, Stack, Queue	1	100
		GRAND TOTAL:	**100 Marks**

Annexure A – Peer Review

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concept	20	Demonstrates complete understanding of the selected data structure/search algorithm with accurate explanations.	Minor omissions but concepts are mostly correct.	Partial understanding with noticeable conceptual gaps.	Unable to explain the concept correctly.
Demonstration	20	Demonstrates all operations correctly using suitable examples or live coding	Demonstrates most operations correctly with minor errors.	Demonstrates limited operations with several errors.	Demonstration is incorrect or incomplete.
Presentation & Communication Skills	20	Organized, confident, clear explanation with effective visual aids.	Good communication with minor presentation issues.	Moderate communication; lacks organization.	Poor communication & presentation.
Question Handling	30	Answers all technical questions confidently with correct reasoning.	Answers most questions correctly.	Answers some questions with partial understanding.	Unable to answer technical questions.
Time Management & Organization	10	Completes presentation within allotted time with smooth flow.	Minor deviation from time.	Significant time management issues.	Presentation poorly managed.

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

Title ---- Browser Back and Forward Navigation Using Stack

Introduction

A web browser allows users to move between previously visited web pages using the **Back** and **Forward** buttons. This navigation is efficiently managed using the **Stack** data structure, which follows the **Last In, First Out (LIFO)** principle. Stack is one of the most common data structures used in real-world applications, and browser navigation is a practical example of its use.

Working Principle

Browser navigation is commonly implemented using **two stacks**:

- **Back Stack:** Stores the history of pages visited. When the user clicks the **Back** button, the current page is removed from the Back Stack, and the browser displays the previous page.
- **Forward Stack:** Stores the pages that the user moves away from after pressing the **Back** button. When the **Forward** button is clicked, the browser returns to the next page by retrieving it from the Forward Stack.

If the user visits a new webpage after going back, the **Forward Stack is cleared**, because a new browsing path has been created.

Advantages

- Provides fast and efficient browser navigation.
- Follows the simple and reliable LIFO principle.
- Improves the user experience by maintaining browsing history.
- Easy to implement using stack operations such as **Push** and **Pop**.

Applications

The stack concept used in browser navigation is also applied in:

- Undo and Redo operations in software.
- Function call management in programming.
- Expression evaluation.
- Navigation history in various applications.

Conclusion

Browser Back and Forward navigation is a practical application of the Stack data structure. By using two stacks, browsers can efficiently manage browsing history and provide smooth navigation. This demonstrates how a simple data structure can solve an important real-world problem efficiently.


oogle → Instragram → Youtube → Github